Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

LESSON PLAN

For

CompTIA Certified A+ Training (Core 1 and Core 2)

TGS Ref No: TGS-2024048317

Conducted by

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

Version 7.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
6.0	1 March 2025	Previous release — 5-day lesson plan covering the nine A+ Core 1 and Core 2 topics.	Dr Alfred Ang
7.1	19 August 2026	Major revision — rebuilt against the current CompTIA A+ Core 1 (220-1101) and Core 2 (220-1102) exam domain weightings. Schedule now covers 54 hands-on labs across the five days, using the browser-based lab toolkit (IP Calculator, PCAP Analyzer, Cybersecurity Simulator, RegexLab) and the Killercoda Ubuntu playground.	Dr Alfred Ang
7.1	19 August 2026	Aligned the Skills Framework section to the full competency set the assessment tests — K1–K6 (Written Assessment) and A1–A8 (Practical Performance) — so no learner is assessed on an outcome the courseware did not declare.	Dr Alfred Ang

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Course Information

Course Title	CompTIA Certified A+ Training (Core 1 and Core 2)
WSQ Course Reference	TGS-2024048317
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
TSC Alignment	Infrastructure Support (ICT-OUS-3007-1.1)
Duration	5 days · 8 training hours per day (40 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Mode	Instructor-led, hands-on IT support labs using physical hardware where available and browser-based tools throughout
Certification Alignment	CompTIA A+ Core 1 (220-1101) and Core 2 (220-1102)
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Diagnose technical issues in network operations and implement procedures to resolve root causes.
- LO2: Troubleshoot technical issues and perform advanced infrastructure configurations.
- LO3: Develop action plans for upgrades and propose improvement ideas based on user needs.
- LO4: Test infrastructure systems and organise information for developing user guides.

Skills Framework Alignment

This course is aligned to the Skills Framework Technical Skill and Competency Infrastructure Support (ICT-OUS-3007-1.1).

Knowledge:

- K1 Diagnostic tools and processes to identify technical issues or disruptions in network infrastructure
- K2 Infrastructure and network configuration techniques
- K3 Troubleshooting techniques for infrastructure technical issues and problems
- K4 Potential benefits and impact of infrastructure upgrades and improvements
- K5 Sources of information for the development of infrastructure user guides and documentation
- K6 Types and purposes of system tests for infrastructure operating requirements

Abilities:

- A1 Diagnose technical issues in network operations
- A2 Implement procedures to resolve root causes of technical issues

- A3 Troubleshoot technical issues in infrastructure systems
- A4 Perform advanced infrastructure configurations
- A5 Develop action plans for infrastructure upgrades
- A6 Propose infrastructure improvement ideas based on user needs
- A7 Test infrastructure systems against operating requirements
- A8 Organise information for the development of user guides

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on support, configuration and troubleshooting tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 5 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Lab Toolkit

All hands-on labs use free, browser-based tools that require no local installation, so every learner works in an identical environment:

Tool	Link	Used for
IP Calculator	https://alfredang.github.io/ipcalculator/	Browser subnet calculator for IPv4 and IPv6 — CIDR, netmask, network and broadcast addresses, usable host ranges and batch processing with CSV export.
PCAP Analyzer	https://alfredang.github.io/pcapanalyzer/	Browser packet-capture analyser — protocol distribution, top talkers, top conversations and a packet table with per-packet detail. Files are parsed locally and never uploaded.
Cybersecurity Simulator	https://alfredang.github.io/cybersecuritysimulator/	Safe threat-simulation lab covering phishing, XSS, SQL injection, password strength, malware, ransomware, social engineering and data leakage.
RegexLab	https://alfredang.github.io/regexgenerator/	Live regular-expression tester with flags, match explanation and substitution — used to filter and parse support logs.
Killercoda Ubuntu Playground	https://killercoda.com/playgrounds/scenario/ubuntu	Free browser-based Ubuntu terminal with root access — no install required. Used for every Linux command-line lab in this course.

Course Schedule

Day 1 — Mobile Devices and Networking

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline and mandatory digital attendance (AM)
10:00–11:00	60 min	**Topic 1 — Mobile Devices: laptop hardware and field-replaceable units, display technologies (LCD/IPS/VA/OLED), digitizers, connection methods, docking (concepts + demo)**
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	Hands-on: Lab 1: Laptop Teardown and Field-Replaceable Unit Identification; Lab 2: Mobile Display Technology and Digitizer Fault Diagnosis; Lab 3: Mobile Connectivity — Bluetooth, NFC, Hotspot and Cellular Configuration; Lab 4: MDM Policy Design and Mobile Synchronisation
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	**Topic 2 — Networking: TCP/UDP, the A+ port list, network devices, wireless standards and bands, SOHO services (concepts + demo). Digital attendance (PM)**
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Hands-on: Lab 5: Protocols and Ports — Building the A+ Port Reference; Lab 6: IPv4 Subnetting and Address Planning with IP Calculator; Lab 7: Network Devices and Traffic Behaviour Analysis
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Hardware, Virtualization and Cloud

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Topic 2 continues — channel planning, SOHO configuration, cabling and packet analysis. Hands-on: Lab 8: Wireless

		Standards, Channel Planning and Interference; Lab 9: SOHO Network Configuration — DHCP, DNS, NAT and Port Forwarding
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	Hands-on: Lab 10: Network Cabling — Termination, Testing and Tool Selection; Lab 11: Packet Capture Analysis — Reading Real Network Traffic. Topic 3 — Hardware: cables and connectors, RAM, storage and RAID (concepts)
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	Hands-on: Lab 12: Cables and Connectors — Identification and Selection; Lab 13: RAM Identification, Installation and Channel Configuration; Lab 14: Storage Devices and RAID Level Selection. Digital attendance (PM)
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Topic 3 continues — motherboards, BIOS/UEFI, CPUs and cooling, power supplies, printers. Hands-on: Lab 15: Motherboard, BIOS/UEFI and CMOS Configuration; Lab 16: CPU Architecture, Sockets and Cooling Solutions
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Hardware and Network Troubleshooting

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 17: Power Supply Sizing, Connectors and Safety; Lab 18: Printer Installation, Configuration and the Laser Imaging Process
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	**Topic 4 — Virtualization and Cloud Computing: hypervisor types, VM resource planning, cloud service and deployment models (concepts). Hands-on: Lab 19: Hypervisor Types and Virtual Machine Resource Planning; Lab 20:**

		Building and Configuring a Linux Virtual Machine; Lab 21: Cloud Service and Deployment Model Selection
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	**Topic 5 — Hardware and Network Troubleshooting: the CompTIA six-step methodology, POST and boot faults (concepts). Hands-on: Lab 22: Applying the CompTIA Six-Step Troubleshooting Methodology; Lab 23: POST, Boot and Power Fault Diagnosis. Digital attendance (PM)**
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Hands-on: Lab 24: Storage and RAID Fault Diagnosis with S.M.A.R.T.; Lab 25: Display and Projector Fault Diagnosis; Lab 26: Mobile Device Hardware Fault Diagnosis
18:15–18:30	15 min	Day 3 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 4 — Operating Systems and Security

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 27: Printer Fault Diagnosis from Print Defects; Lab 28: Network Fault Diagnosis — Command Line to Packet Level. End of Core 1 domains
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	**Topic 6 — Operating Systems: Windows editions and installation, partitioning, command line, Windows tools (concepts). Hands-on: Lab 29: Windows Editions, Installation Types and Partitioning; Lab 30: Windows Command Line — Navigation, Files and Copy Operations; Lab 31: Windows Networking and Repair Commands**
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	Hands-on: Lab 32: Windows Management Tools — Task Manager, MMC and Snap-ins; Lab 33: File Systems,

		Permissions and Share Configuration; Lab 34: Linux Command Line Essentials on Killercoda. Digital attendance (PM)
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Hands-on: Lab 35: Linux Permissions, Ownership and User Management; Lab 36: macOS Tools and Cross-Platform File System Compatibility. Topic 7 — Security: physical and logical controls, authentication (concepts). Hands-on: Lab 37: Physical and Logical Security Control Design
18:15–18:30	15 min	Day 4 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 5 — Software Troubleshooting, Operational Procedures and Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Topic 7 continues — malware, social engineering, wireless and workstation hardening. Hands-on: Lab 38: Password Strength Analysis and Authentication Policy; Lab 39: Phishing Recognition and Social Engineering Defence; Lab 40: Malware Types, Symptoms and the Seven-Step Removal Procedure
11:00–11:15	15 min	Tea break
11:15–12:45	90 min	Hands-on: Lab 41: Network Attack Recognition — Injection, XSS and Data Leakage; Lab 42: Wireless Security and SOHO Router Hardening; Lab 43: Windows Security Configuration and BitLocker Encryption; Lab 44: Data Destruction, Disposal and Regulated Data Handling
12:45–13:30	45 min	**Topic 8 — Software Troubleshooting: Windows symptoms and recovery tools, malware and browser symptoms (concepts). Hands-on: Lab 45: Windows OS Symptom Diagnosis and Recovery Tools**
13:30–14:30	60 min	Lunch break
14:30–15:45	75 min	Hands-on: Lab 46: Log Analysis with Regular

		Expressions; Lab 47: Malware Symptom Response and Browser Problem Resolution; Lab 48: Mobile OS and Application Troubleshooting; Lab 49: System Restore, Backup Verification and Recovery Testing. Digital attendance (PM)
15:45–16:15	30 min	**Topic 9 — Operational Procedures: documentation, change and asset management, backup, safety, professionalism. Hands-on: Lab 50: Ticketing, Documentation and Knowledge Base Authoring; Lab 51: Change Management and Asset Management; Lab 52: Backup Strategy Design and the 3-2-1 Rule; Lab 53: Safety Procedures, ESD Control and Environmental Compliance; Lab 54: Professional Communication, Scripting and Remote Access**
16:15–16:30	15 min	**Course recap, exam preparation guidance, TRAQOM survey and Briefing for Assessment**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — hands-on support, configuration and troubleshooting tasks, 1 hour, open book. Assessment digital attendance**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to CompTIA A+ exam domains)

Topic / Exam domain	Exam	Exam weighting	Course time	Labs
Topic 01: Mobile Devices	Core 1	13% of Core 1	7%	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Networking	Core 1	23% of Core 1	12%	Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11
Topic 03: Hardware	Core 1	25% of Core 1	13%	Lab 12, Lab 13, Lab 14, Lab 15, Lab 16, Lab 17, Lab 18
Topic 04:	Core 1	11% of Core 1	6%	Lab 19, Lab 20,

Virtualization and Cloud Computing				Lab 21
Topic 05: Hardware and Network Troubleshooting	Core 1	28% of Core 1	15%	Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27, Lab 28
Topic 06: Operating Systems	Core 2	28% of Core 2	17%	Lab 29, Lab 30, Lab 31, Lab 32, Lab 33, Lab 34, Lab 35, Lab 36
Topic 07: Security	Core 2	28% of Core 2	17%	Lab 37, Lab 38, Lab 39, Lab 40, Lab 41, Lab 42, Lab 43, Lab 44
Topic 08: Software Troubleshooting	Core 2	23% of Core 2	8%	Lab 45, Lab 46, Lab 47, Lab 48, Lab 49
Topic 09: Operational Procedures	Core 2	21% of Core 2	5%	Lab 50, Lab 51, Lab 52, Lab 53, Lab 54